



ANALYZING BOOKING TRENDS IN THE HOSPITALITY INDUSTRY

A predictive model to determine the booking status of a customer



-- Business Introduction --

Hotel Haven is a luxury hotel chain with multiple locations. The hotel offers a wide range of services, including **various types of rooms, meal plans, and parking options.**

Hotel Haven has been facing challenges in **predicting customer cancellations**, which impacts their ability to **manage resources** effectively. The hotel wants to better **understand their booking patterns** and **create strategies** that could **improve customer retention** and **reduce cancellations.**

-- PROBLEM STATEMENT --

The hotel struggles with **high cancellation rates**, leading to **lost revenue** and **inefficient resource allocation**. The existing system does not provide sufficient insights into why customers cancel their bookings. The hotel management seeks to **predict cancellations** based on **booking data** to **improve operational efficiency** and **customer retention**.



-- Rationale for the

• **Understanding Cancellation** **Project—**

- Analyzing booking patterns helps uncover the underlying reasons why customers cancel their bookings.

• **Identifying Key Factors**

- By examining variables like lead time, room type, market segment, and price, the project allows the identification of specific factors that influence customers' decisions to either cancel or keep their bookings.

• **Optimizing customer Satisfaction**

- By minimizing cancellations, the hotel can ensure better resource allocation and improved guest experience. Customers who are less likely to cancel will contribute to smoother operations and increased



--Deliverables--

Data Cleaning

Data Explorations

Comprehensive Data
Analysis

Feature Engineering
and Model
Development

Model Evaluation and
Fine Tuning

Jupyter Notebook
Powerpoint Slides

Column Name	Description	Data Type
Booking_ID	Unique identifier for each booking	String
number of adults	Number of adults in the booking	Integer
number of children	Number of children in the booking	Integer
number of weekend nights	Number of weekend nights included in the booking	Integer
number of week nights	Number of week nights included in the booking	Integer
type of meal	Meal plan selection (e.g., Meal Plan 1, Not Selected)	Categorical
car parking space	Whether the booking includes parking (0: No, 1: Yes)	Binary
room type	Type of room booked (e.g., Room_Type 1, Room_Type 2)	Categorical
lead time	Time between the reservation and check-in date	Integer
market segment type	Booking channel (e.g., Online, Offline)	Categorical

Column Name	Description	Data Type
repeated	Whether the booking is from a repeat customer (0: No, 1: Yes)	Binary
P-C	Customer profile type indicator (0: Not Customer, 1: Customer)	Binary
P-not-C	Non-customer profile type indicator (0: Not Non-Customer, 1: Non-Customer)	Binary
average price	Average price of the booking	Float
special requests	Number of special requests made by the customer	Integer
date of reservation	Date the reservation was made	Date
booking status	Status of the booking (e.g., Canceled, Not Canceled)	Categorical